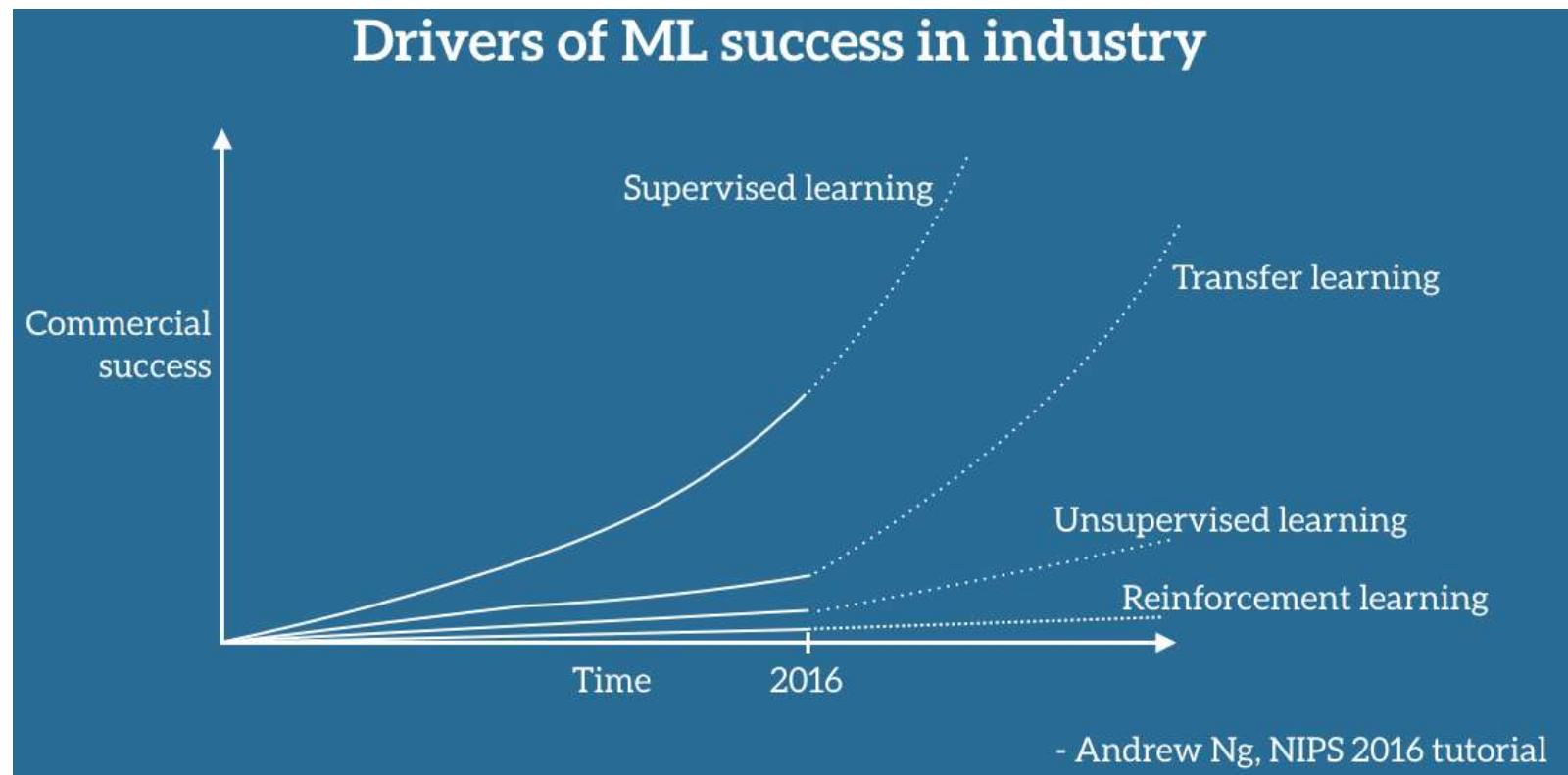


Transfer Learning

- Humans have an inherent ability to transfer knowledge across tasks.
- What we acquire as knowledge while learning about one task, we utilize in the same way to solve related tasks.
- The more related the tasks, the easier it is for us to transfer, or cross-utilize our knowledge. Some simple examples would be:
 - now how to ride a motorbike ➡ Learn how to ride a car
 - Know how to play classic piano ➡ Learn how to play jazz piano
 - Know math and statistics ➡ Learn machine learning

Motivation of Transfer Learning



Drivers of ML industrial success according to Andrew Ng

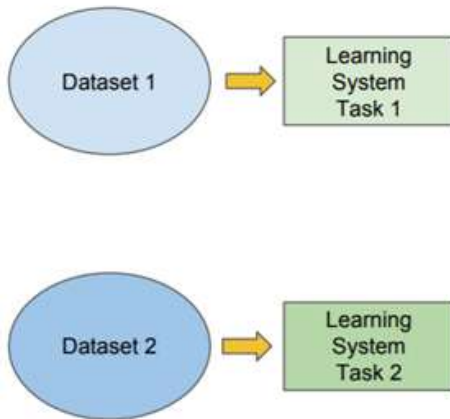
Transfer Learning

- In their famous book, *Deep Learning*, Goodfellow et al. refer to transfer learning in the context of generalization. Their definition is as follows: “*Situation where what has been learned in one setting is exploited to improve generalization in another setting.*”
- **Transfer learning (TL)** is a research problem in machine learning (ML) that focuses on storing knowledge gained while solving one problem and applying it to a different but related problem. For example, knowledge gained while learning to recognize cars could apply when trying to recognize trucks.

Traditioanal ML vs Transfer Learning

Traditional ML

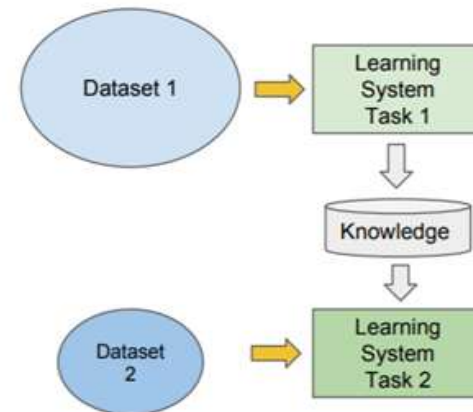
- Isolated, single task learning:
 - Knowledge is not retained or accumulated. Learning is performed w.o. considering past learned knowledge in other tasks



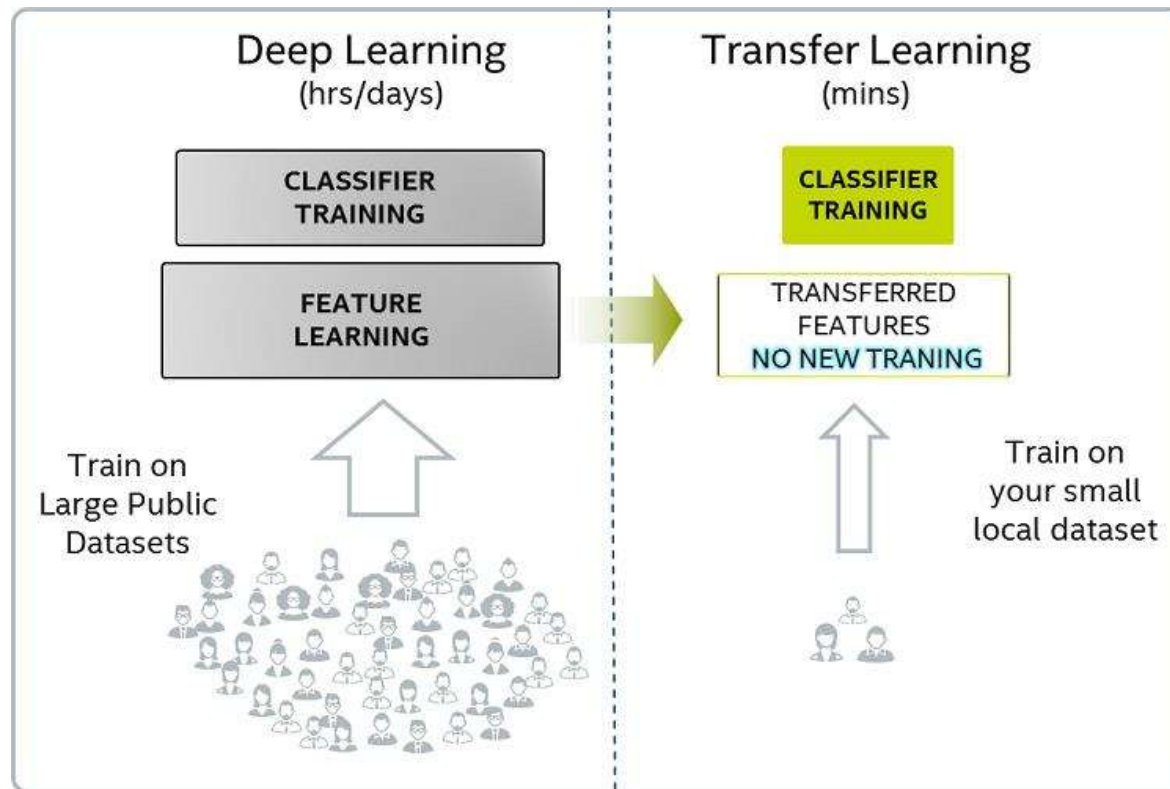
vs

Transfer Learning

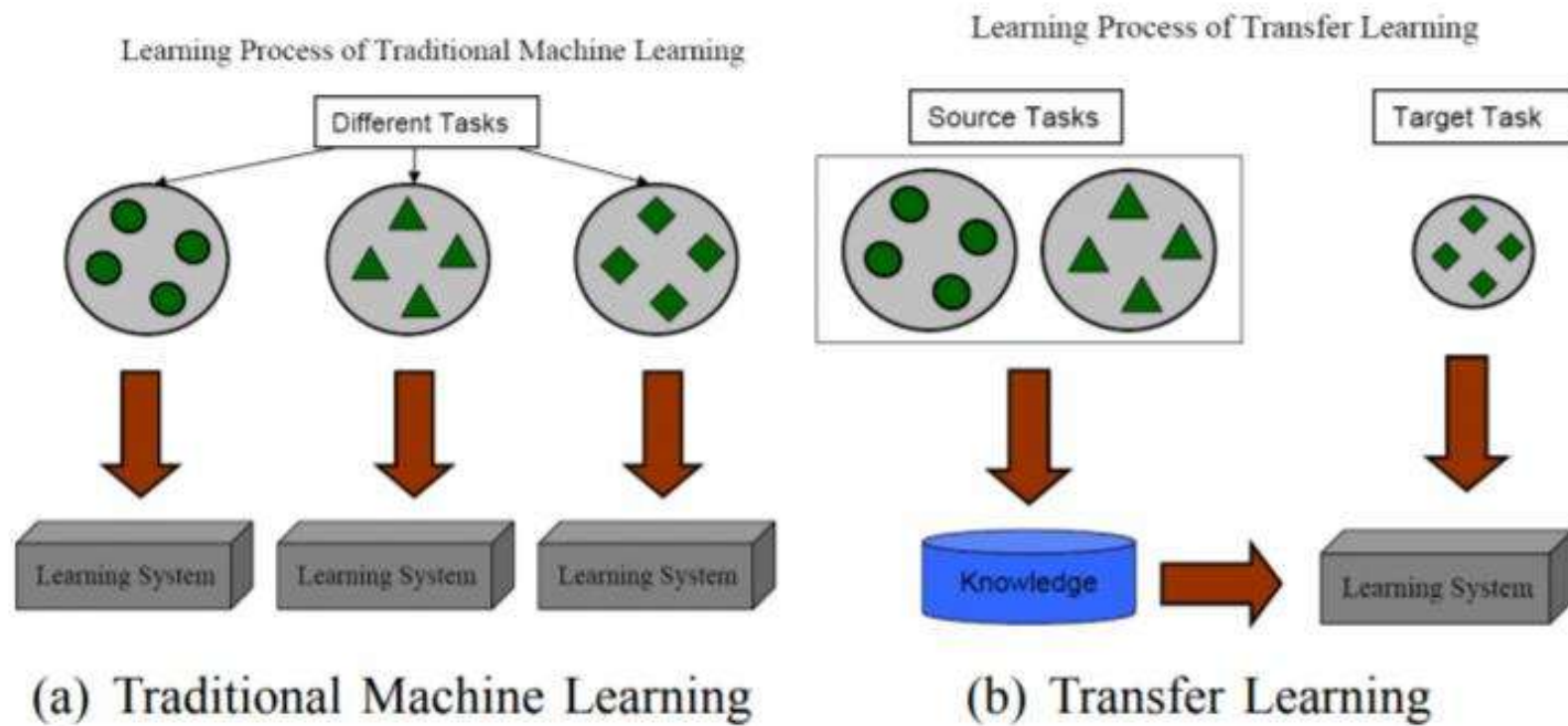
- Learning of a new tasks relies on the previous learned tasks:
 - Learning process can be faster, more accurate and/or need less training data



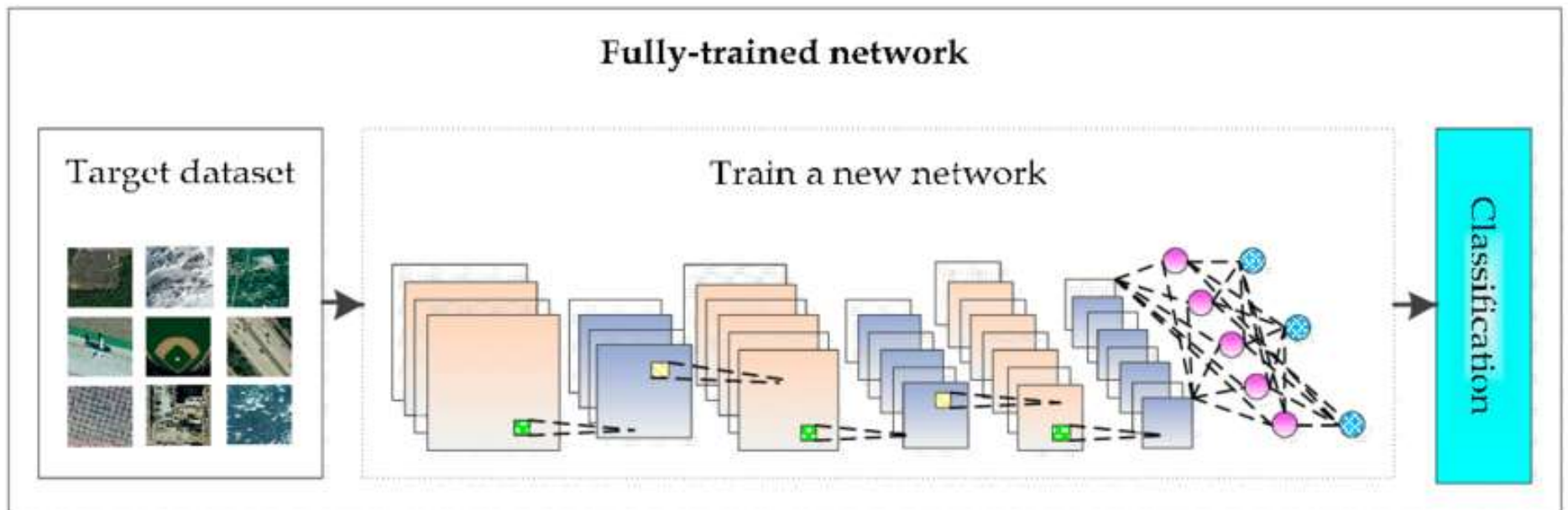
Why Transfer Learning?



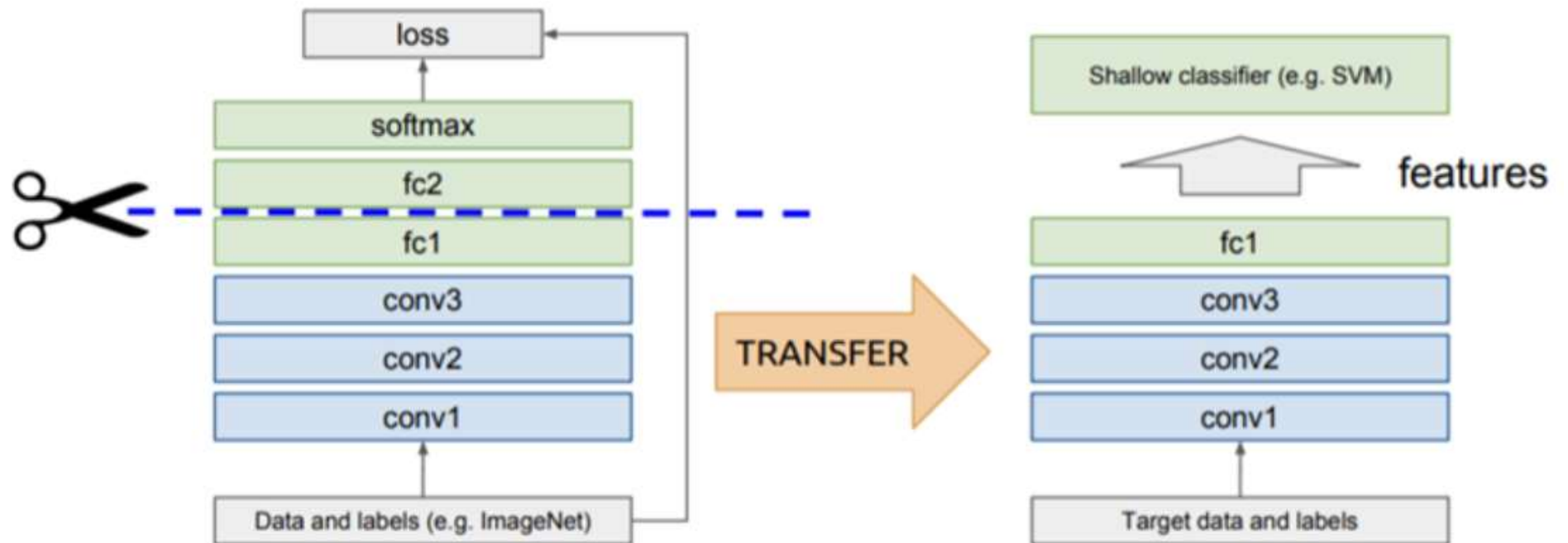
Transfer Learning



Fully Trained Network

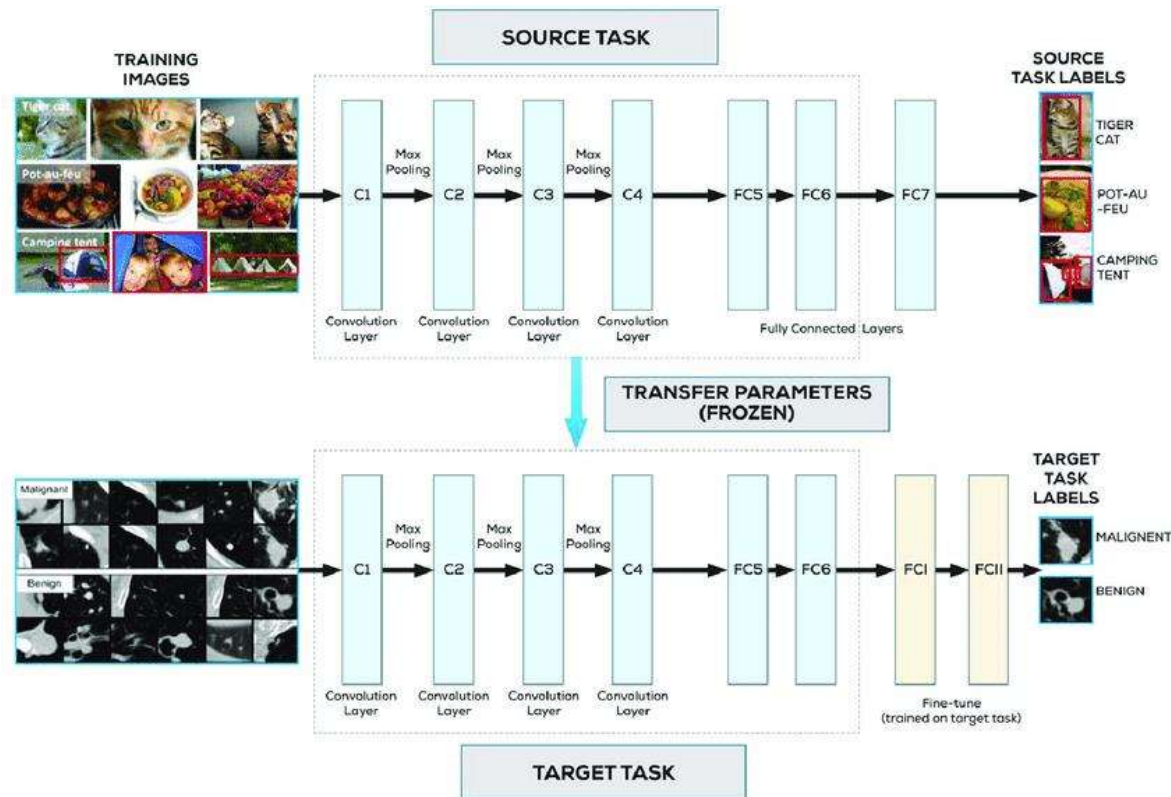


Transfer Learning



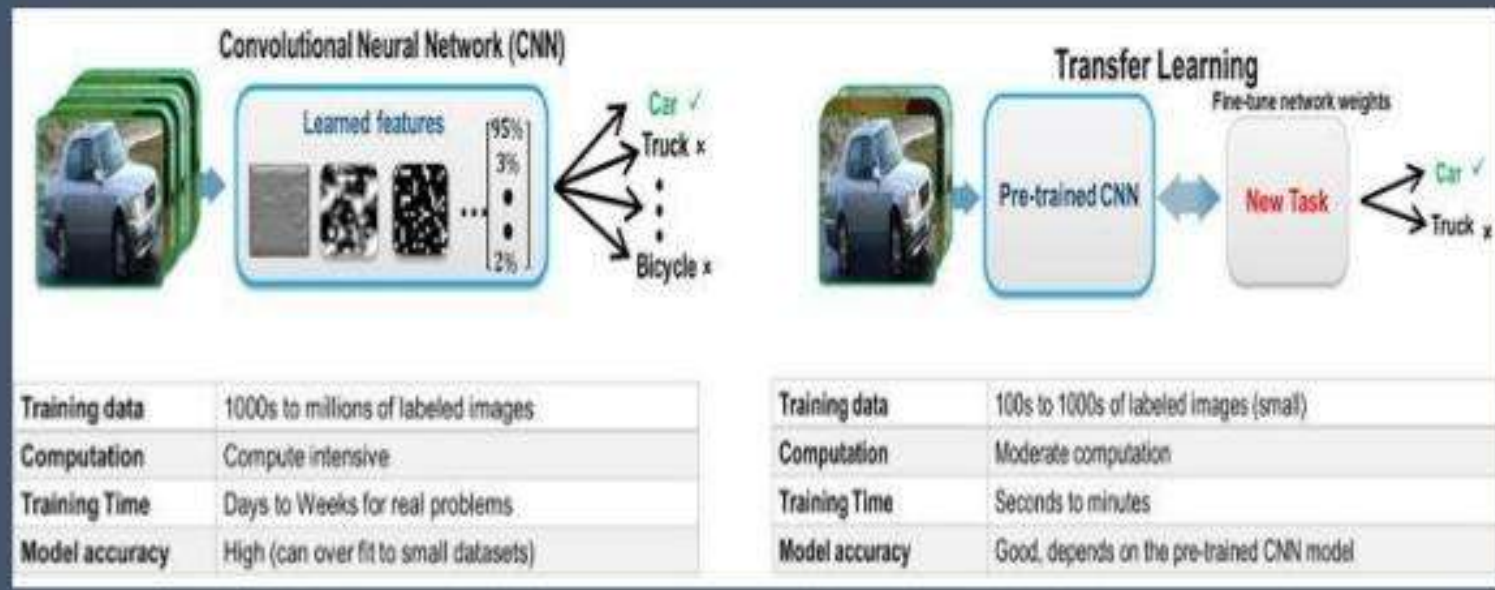
Transfer Learning

- There are different options, which layers should be frozen and which should be retrained or fine tuned.

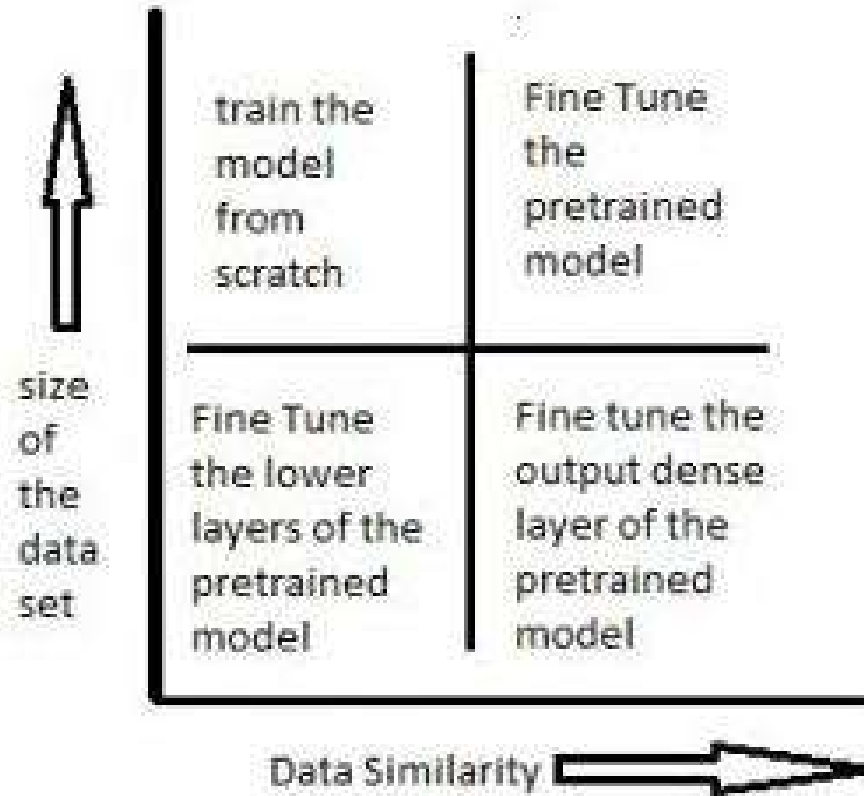


Transfer Learning

Transfer Learning & Fine-tuning DCNN



General strategy



Transfer Learning: Controlling layers which need to be frozen

